

Research & Report (Theory Task) (Due by Feb 3 11:59pm)

Each member must research and prepare a document (1-2 pages) explaining:

Types of malware and their impact.

A famous malware attack (e.g., WannaCry, Stuxnet, Pegasus) and how it worked.

Submission: push your documented pdf as week 3 day 1 by today itself .

Malware

- **Definition:** Malware, short for malicious software, refers to any software intentionally designed to cause damage to a computer, server, client, or network. It can damage, or disable computers and systems.

The types of Malware and its impact are following :

1. Viruses

- **Definition:** Malicious programs that attach themselves to legitimate files or programs and replicate when those programs are run.
- **Impact:** Can corrupt, delete, or steal data, and disrupt system operations. They often spread to other files and systems, causing widespread damage.

2. Worms

- **Definition:** Self-replicating malware that spreads across networks by exploiting vulnerabilities, without needing any user interaction.
- **Impact:** Can cause network congestion, slow down systems, and create openings for additional malware, leading to further infection and damage.

3. Trojan Horses

- **Definition:** Malicious software that looks as legitimate, useful programs to trick users into installing them.
- **Impact:** Can steal sensitive information, install other types of malware, and grant remote access to attackers, compromising system security and data.

4. Ransomware

- **Definition:** Malware that encrypts files or locks users out of their systems, demanding a ransom to restore access.
- **Impact:** Causes significant financial loss as victims are pressured to pay to regain access to their files or systems. It can lead to data inaccessibility and disrupt operations.

5. Spyware

- **Definition:** Software that secretly monitors and collects user information and activities without their knowledge.
- **Impact:** Leads to privacy breaches and potential identity theft by gathering sensitive information such as passwords, credit card numbers, and personal details.

6. Adware

- **Definition:** Software that automatically displays or downloads advertisements when a user is online.
- **Impact:** Slows down system performance, bombards users with unwanted ads, and can lead to further malware infections by redirecting users to malicious websites.

7. Rootkits

- **Definition:** Stealthy software that hides within a system and provides privileged access to the attacker.
- **Impact:** Enables attackers to maintain hidden control over a system, protect other malware, and perform unauthorized actions without detection.

8. Keyloggers

- **Definition:** Programs that record every keystroke made by a user.
- **Impact:** Capture sensitive information such as usernames, passwords, and credit card numbers, which can lead to data breaches, financial theft, and identity theft.

9. Botnets

- **Definition:** Networks of infected computers, or "bots," controlled remotely by attackers to perform large-scale malicious activities.
- **Impact:** Used to launch Distributed Denial of Service (DDoS) attacks, send spam emails, steal data, and carry out other malicious activities on a large scale.

10. Fileless Malware

- **Definition:** Malware that operates in the computer's memory rather than from files stored on the hard drive.
- **Impact:** Harder to detect as it doesn't leave traditional traces on the hard drive. It exploits system vulnerabilities to carry out malicious activities and evade security measures.

WannaCry Ransomware Attack

Overview: The WannaCry ransomware attack occurred in May 2017 and was one of the most widespread and damaging cyberattacks in history. It targeted computers running the Microsoft

Windows operating system by encrypting data and demanding ransom payments in Bitcoin to restore access.

How it Worked:

1. **Exploitation of Vulnerability:** WannaCry exploited a vulnerability in the Windows implementation of the Server Message Block (SMB) protocol, known as "EternalBlue." This exploit was originally developed by the United States National Security Agency (NSA) and leaked by a group called The Shadow Brokers.
2. **Propagation:** Once a computer was infected, WannaCry would attempt to spread to other vulnerable machines on the same network. It used a combination of worm-like behavior and ransomware, making it highly contagious.
3. **Encryption:** The malware encrypted files on the infected computer, making them inaccessible to the user. It targeted a wide range of file types, including documents, images, and videos³.
4. **Ransom Demand:** After encrypting the files, WannaCry displayed a ransom note demanding payment of \$300 to \$600 in Bitcoin to decrypt the files. If the ransom was not paid within a specified time frame, the ransom amount would double.
5. **Kill Switch:** Interestingly, a security researcher discovered a "kill switch" in the malware's code, which temporarily halted the spread of WannaCry. However, many systems remained infected and required manual intervention to remove the malware.

Impact: The attack affected over 300,000 computers across 150 countries, causing significant disruption to organizations, including hospitals, government agencies, and businesses. The estimated financial damage ranged from hundreds of millions to billions of dollars.

Stuxnet Malware Attack

Overview: Stuxnet is a sophisticated computer worm discovered in 2010, believed to be a joint effort by the United States and Israel to target Iran's nuclear program. It was designed to sabotage Iran's uranium enrichment by causing physical damage to the centrifuges used in the process.

How it Worked:

1. **Infection:** Stuxnet initially spread through infected USB flash drives, exploiting multiple zero-day vulnerabilities in the Windows operating system.
2. **Targeting:** Once inside a system, Stuxnet searched for Siemens Step7 software, which is used to program industrial control systems.
3. **Sabotage:** Stuxnet altered the code in the PLCs (Programmable Logic Controllers) to make the centrifuges spin at erratic speeds, causing physical damage while showing normal operation to the monitoring systems.
4. **Stealth:** The worm used rootkit techniques to hide its presence and avoid detection.

Impact: Stuxnet reportedly destroyed nearly one-fifth of Iran's nuclear centrifuges, causing significant setbacks to their nuclear program. It also highlighted the potential for cyberweapons to cause physical damage and raised concerns about the vulnerability of critical infrastructure to cyberattacks.

Pegasus Spyware Attack

Overview: Pegasus is a highly sophisticated spyware developed by the Israeli cybersecurity firm NSO Group. It is designed to stealthily infiltrate smartphones and other devices to gather information and monitor activities. Pegasus first gained widespread attention in 2016 when it was discovered being used to target human rights activists and journalists.

How it Worked:

1. **Infection:** Pegasus can be delivered through various methods, including malicious links, text messages, and zero-click exploits that require no user interaction. It exploits vulnerabilities in operating systems like iOS and Android to gain access.
2. **Capabilities:** Once installed, Pegasus can take complete control of the infected device. It can access messages, emails, photos, and contacts, record calls, activate the microphone and camera, track the device's location, and monitor internet activity.
3. **Stealth:** Pegasus is designed to operate covertly, avoiding detection by traditional antivirus and security software. It can also delete itself to remove traces of the attack.

Impact: The discovery of Pegasus has raised significant concerns about privacy and surveillance. It has been used by governments and other entities to spy on activists, journalists, politicians, and other high-profile individuals. The widespread use of Pegasus has led to legal and ethical debates about the balance between security and privacy.